AUTHOR'S PREFACE

All physicists agree that the problem of physics consists in tracing the phenomena of nature back to the simple laws of mechanics. But there is not the same agreement as to what these simple laws are. To most physicists they are simply Newton's laws of motion. But in reality these latter laws only obtain their inner significance and their physical meaning through the tacit assumption that the forces of which they speak are of a simple nature and possess simple properties. But we have here no certainty as to what is simple and permissible, and what is not: it is just here that we no longer find any general agreement. Hence there arise actual differences of opinion as to whether this or that assumption is in accordance with the usual system of mechanics, or not. It is in the treatment of new problems that we recognise the existence of such open questions as a real bar to progress. So, for example, it is premature to attempt to base the equations of motion of the ether upon the laws of mechanics until we have obtained a perfect agreement as to what is understood by this name.

The problem which I have endeavoured to solve in the present investigation is the following:—To fill up the existing gaps and to give a complete and definite presentation of the laws of mechanics which shall be consistent with the state of our present knowledge, being neither too restricted nor too extensive in relation to the scope of this knowledge. The presentation must not be too restricted: there must be no natural motion which it does not embrace. On the other

hand it must not be too extensive: it must admit of no motion whose occurrence in nature is excluded by the state of our present knowledge. Whether the presentation here given as the solution of this problem is the only possible one, or whether there are other and perhaps better possible ones, remains open. But that the presentation given is in every respect a possible one, I prove by developing its consequences, and showing that when fully unfolded it is capable of embracing the whole content of ordinary mechanics, so far as the latter relates only to the actual forces and connections of nature, and is not regarded as a field for mathematical exercises.

In the process of this development a theoretical discussion has grown into a treatise which contains a complete survey of all the more important general propositions in dynamics, and which may serve as a systematic text-book of this science. For several reasons it is not well suited for use as a first introduction; but for these very reasons it is the better suited to guide those who have already a fair mastery of mechanics as usually taught. It may lead them to a vantage-ground from which they can more clearly perceive the physical meaning of mechanical principles, how they are related to each other, and how far they hold good; from which the ideas of force and the other fundamental ideas of mechanics appear stripped of the last remnant of obscurity.

In his papers on the principle of least action and on cyclical systems,[1] von Helmholtz has already treated in an indirect manner the problem which is investigated in this book, and has given one possible solution of it. In the first set of papers he propounds and maintains the thesis that a system of mechanics which regards as of universal validity, not only Newton's laws, but also the special assumptions involved (in addition to these laws) in Hamilton's Principle,

[1] H. von Helmholtz, "Über die physikalische Bedeutung des Prinzips der kleinsten Wirkung," *Journal für die reine und angewandte Mathematik*, **100**, pp. 137-166, 213-222, 1887; "Prinzipien der Statik monocyklischer Systeme," *ibid.* **97**, pp. 111-140, 317-336, 1884.

would yet be able to embrace all the processes of nature. In the second set of papers the meaning and importance of concealed motions is for the first time treated in a general way. Both in its broad features and in its details my own investigation owes much to the above-mentioned papers: the chapter on cyclical systems is taken almost directly from them. Apart from matters of form, my own solution differs from that of von Helmholtz chiefly in two respects. Firstly, I endeavour from the start to keep the elements of mechanics free from that which von Helmholtz only removes by subsequent restriction from the mechanics previously developed. Secondly, in a certain sense I eliminate less from mechanics, inasmuch as I do not rely upon Hamilton's Principle or any other integral principle. The reasons for this and the consequences which arise from it are made clear in the book itself.

In his important paper on the physical applications of dynamics, J. J. Thomson[1] pursues a train of thought similar to that contained in von Helmholtz's papers. Here again the author develops the consequences of a system of dynamics based upon Newton's laws of motion and also upon other special assumptions which are not explicitly stated. I might have derived assistance from this paper as well; but as a matter of fact my own investigation had made considerable progress by the time I became familiar with it. I may say the same of the mathematical papers of Beltrami[2] and Lipschitz,[3] although these are of much older date. Still I found these very suggestive, as also the more recent presentation of their investigations which Darboux[4] has given with

[1] J. J. Thomson, "On some Applications of Dynamical Principles to Physical Phenomena," *Philosophical Transactions*, **176**, II., pp. 307-342, 1885.

[2] Beltrami, "Sulla teoria generale dei parametri differenziali," *Memorie della Reale Accademia di Bologna*, 25 Febbrajo 1869.

[3] R. Lipschitz, "Untersuchungen eines Problems der Variationsrechnung, in welchem das Problem der Mechanik enthalten ist," *Journal für die reine und angewandte Mathematik*, **74**, pp. 116-149, 1872. "Bemerkungen zu dem Princip des kleinsten Zwanges," *ibid.* **82**, pp. 316-342, 1877.

[4] G. Darboux, *Leçons sur la théorie générale des surfaces*, livre v., chapitres vi. vii. viii., Paris, 1889.